Project Title : *Rainfall Prediction*

Subject : *Intro to Machine Learning (BCSC 0712)*

Team name : *Forecast Force*

Team Members : *Manikanta ,Nikhil , Harini*

Guide : *Dr.Shrawan Kumar*
proffesor of practice
SAP AI HEAD and EX-scientist
Dept.of CSE

GLA UNIVERSITY , GREATER NOIDA

Content

- *INTRODUCTION*
- ***Dataset & Features***
- ***Machine Learning Model***
- ***Algorithms that can be used***
- **Conclusion**

Introduction

Rainfall Prediction is the process of forecasting rainfall using historical weather data.

Why it is Important :

Helps farmers plan crops
Prevents flood damage 🌧️
Helps in water resource management
Useful for weather forecasting

Objective of the Project:

To build a Machine Learning model that predicts rainfall based on weather parameters.

Dataset & Features

The dataset contains historical weather information.

Important features used

- Temperature
- Humidity
- Wind Speed
- Atmospheric Pressure
- Cloud Cover
- Previous Rainfall Data

These features help the ML model learn patterns and predict rainfall.

Machine Learning Model

Steps used in the project:

- *Data Collection*
- *Data Preprocessing (cleaning missing values)*
- *Feature Selection*
- *Model Training*

Algorithms that can be used:

- *Linear Regression*
- *Decision Tree*
- *Random Forest*
- *Support Vector Machine*

The trained model predicts whether rainfall will occur or not.

Conclusion

Rainfall prediction using Machine Learning improves weather forecasting accuracy.

Advantages:

Faster predictions
Better accuracy
Helps agriculture and disaster management

Future Scope:

Use larger datasets
Apply Deep Learning models
Improve prediction accuracy

Rainfall prediction using Machine Learning helps improve the accuracy of weather forecasting by analyzing historical weather data.
It is useful for agriculture planning, disaster management, and efficient water resource management.